Learning Problem

$E_{out}(g^*) \leq E_{in}(g^*) +$ Generalization Bound

- Minimize $E_{in}(g^*)$ and the Generalization Bound

$E_{out}(g^*) \vdash$ Generalization Bound $\dashv E_{in}(g^*)$

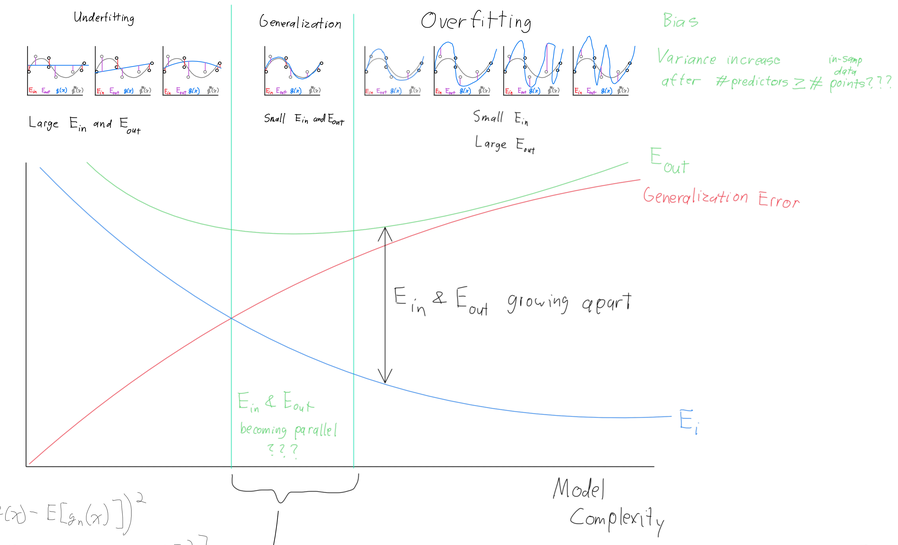
minimize $E_{in}(g^*)$

$E_{out}(g^*) \vdash$ Generalization Bound $\dashv E_{in}(g^*)$

Minimize General Bound

$E_{out}(g^*) \vdash E_{in}(g^*)$

Generalization Bound

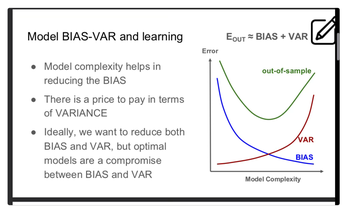


Bias

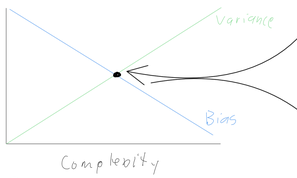
Variance increase after # predictors ≥ # in-sample data points???

$Bias = \left(f(x) - E[g_n(x)]\right)^2$

$Var = E\left[\left(g_n(x) - E[g_n(x)]\right)^2\right]$



- Model complexity helps in reducing the BIAS
- There is a price to pay in terms of VARIANCE
- Ideally, we want to reduce both BIAS and VAR, but optimal models are a compromise between BIAS and VAR



reduce model complexity to reduce the variance to to the bias/variance sweet spot

Increase the in-sample data to reduce variance w/o increasing bias

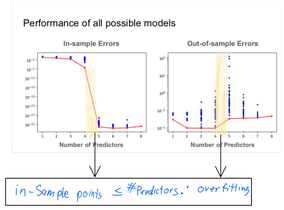
Bias, Variance Tradeoff

Overcoming Over Fitting

Subset Selection

Find what subset of predictors better (fits??) explains target

Best$(\min(E_{out}))$ of all $\left(\sum_k^n C_k^n\right)$ trained model subsets



in-Sample points ≤ #Predictors ∴ overfitting

Stepwise Selection

Explore Restricted set of models

Forward: From null model, add on predictor at a time

$y_1 = 0$

$y_2 = w_0$

$y_3 = w_0 + w_1 x$

⋮

$y_n = w_0 + w_1 x + \ldots + w_n x_n^n$

Backward: Iteratively remove one predictor from full Model

$y_n = w_0 + w_1 x + \ldots + w_n x_n^n$

$y_{n-1} = w_0 + w_1 x + \ldots + w_{n-1} x_{n-1}^{n-1}$

⋮

$y_3 = w_0 + w_1 x$

$y_2 = w_0$

$y_1 = 0$

Mixed: Doing both

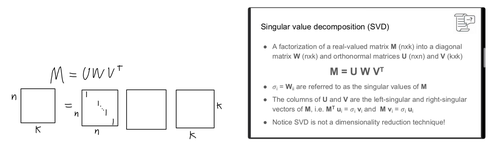
Dimensionality Reduction

- Transform the set of predictors $\{p_k\}$ in to smaller better subset of predictors $\{z_m\}$

$$Z_m = \sum_{k=1}^{K} \phi_{km} P_k \qquad M \ll k$$

$$\{z_0 \ldots z_m \ldots z_M\}$$

SVD Not dimensionality



PCA

$T = MV = UW$ where $M = UWV^T$

Reduces Noise and Concentrated

Why models overfit?

- Stochastic Noise
- Deterministic Noise
- # in-sample datapoints

} Model will Start fitting these

$$E\left[\left(f(x) + \varepsilon - g(x)\right)^2\right] = \underbrace{\left(f(x) - \bar{g}(x)\right)^2}_{\text{Bias}} + \underbrace{E\left[\left(g(x) - \bar{g}(x)\right)^2\right]}_{\text{var}} + \underbrace{var(\varepsilon)}_{\text{Irreducible}}$$

Deterministic Noise
- Part of target function not properly fit by the model

Stochastic Noise
- Inherent Noise from the data

How does Deterministic noise cause overfitting???

Stochastic mainly from Irreducible and partly from model variance

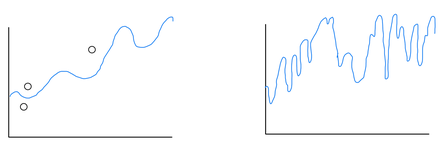
or

Stochastic affects model variance and Irreducible

↑ Stochastic Noise Overfitting

$MSE = Bias + var + var(\varepsilon)$

Case study

10th order target + noise

Noiseless 50th-order target